

Business problem

↓

Data

↓

Statistics

↓

programming

↓

feature engineering

↓

Machine Learning

↓

Deep Learning

Generative AI

↓

System design

↓

deployment

↓

monitoring

Phase 7: Monitoring & Evaluation ML

→ Data drift, model drift, performance monitoring, retraining, model versioning, rollback

End to End ML project

Phase 8: Deep Learning

→ Neural Networks

↳ Perceptron, MLP, Forward pass, Loss, backprop, activation functions.

→ CNN

↳ VGG, ResNet

→ RNN/LSTM/GRU

→ TensorFlow / PyTorch

AI → Artificial Intelligence

↳ Traditional AI

↳ Machine Learning

↳ Deep Learning

↳ Gen AI

ML: Machine Learning

↳ Supervised

↳ Unsupervised

↳ Reinforcement ML

Syllabus

Phase 1: Python

→ Variables, data types, Operators, Control flow, functions, Data structures (list, set, tuple, dictionary), file handling, OOPS (Object oriented programming)

Phase 2: Statistics

→ mean, median, mode, variance, std dev, normal distribution, binomial distribution, poisson distribution, PDF, CDF, z score

→ Inference → Hypothesis testing, z test, t test, chi square test, p value and p value interval,

→ Advance testing → Bayesian inference, Simpson's paradox, sample - sz calculation, sampling techniques.

→ Pearson correlation, Spearman correlation, covariance, regression, confounding variables, interaction effects

Phase 3: Mathematics

→ Calculus → Derivatives, chain rules, gradients, loss functions, Gradient Descent, Convexity

→ Linear Algebra → Vectors, Matrices, Matrix ops, eigenvalues, eigenvectors, matrix decomposition, linear transformation

→ Probability & Bayesian method

→ Time Series problems.

Phase 4: Libraries

→ Numpy + Pandas

→ Visualization: Matplotlib + Seaborn + Plotly

→ SQL session

Phase 5: Machine Learning

ML-I Fundamentals

→ ML pipeline, SK, VL, RL, train/validation/test, cross validation, Stick-it-learn workflow.

ML-II - Regression

→ Linear regression, cost function, GD, L1, L2, ElasticNet, feature scaling

ML-III - Classification

→ Logistic regression, decision boundaries, multiclass, softmax

ML-IV - Tree models

→ Decision tree, Random forest, Gradient boosting, XGBoost, LightGBM, Catboost, Explainable AI (LIME + SHAP), hyperparameters tuning

ML-V → Advanced Algorithms

→ SVM, KNN, Naive Bayes, DBSCAN, PCA, t-SNE, UMAP

Phase 6: Recommendation Systems

→ Content based, collaborative filtering, hybrid

Phase 9 → NLP & Transformers

→ Text preprocessing, word2vec, glove, transformer, self attention, BERT, GPT basics.

Phase 10 → Computer vision

↳ Image preprocessing, object detection, image captioning

Phase 11: Gen AI

↳ LLM fundamentals

↳ LLM architecture

↳ tokenization

↳ embeddings

↳ context window

↳ temperature & sampling

↳ instruction tuning

↳ Prompt Engineering

↳ Zero-shot

↳ few shot

↳ Chain of thoughts

↳ prompt templates

↳ prompt optimization

↳ prompt caching

RAG

↳ FAISS, Pinecone, weaviate, milvus

Agents + multi agent + multimodal

↳ function calling

Phase 12: ML Systems: Model tuning & optimization.

GDPR & CCPA

Capstone 1: Enterprise search & Ranking system.

Capstone 2: Real-time personalization platform

AI/ML

↳ Data Science

↳ Python + SQL

↳ Stats

↳ Pandas

↳ ML

↳ AI Engineering

↳ LLMs

↳ RAG

↳ Agent

↳ Multimodal

↳ ML Systems

↳ Serving & MLOps

↳ Capstone